Tutorial - 6.   10/8/26

Ridge regression (linear model)

Bedroom (x_1)	Age (years) (x_2)	Price (y)
3	5	2
3	3	3

linear eq^n.

$$2 = c + 3m_1 + 5m_2$$
$$3 = c + 3m_1 + 3m_2$$

$$\underset{y}{\begin{pmatrix} 2 \\ 3 \end{pmatrix}} = \underbrace{\begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix}}_{x} \underset{w}{\begin{pmatrix} c \\ m_1 \\ m_2 \end{pmatrix}}$$

$$w = (x^T x + \lambda I)^{-1} (x^T y)$$

Take $\lambda = 1$.

$$x = \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix} \quad x^T = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix}$$

$$x^T x = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 1 & 3 & 5 \\ 1 & 3 & 3 \end{pmatrix} = \begin{pmatrix} 1+1 & 3+3 & 5+3 \\ 3+3 & 9+9 & 15+9 \\ 5+3 & 15+9 & 25+9 \end{pmatrix}$$

$$x^T x = \begin{pmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{pmatrix}$$

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$(x^T x + \lambda I)$$

$$= \begin{pmatrix} 2 & 6 & 8 \\ 6 & 18 & 24 \\ 8 & 24 & 34 \end{pmatrix} + 1 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 3 & 6 & 8 \\ 6 & 19 & 24 \\ 8 & 24 & 34 \end{pmatrix}$$

$$(X^T X + \lambda I)^{-1} = \frac{\text{Adj}\,(X^T X + \lambda I)^{-1}}{\det(X^T X + \lambda I)^{-1}}$$

$$= \frac{1}{95} \begin{bmatrix} 89 & -18 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{bmatrix}$$

$$X^T Y = \begin{pmatrix} 1 & 1 \\ 3 & 3 \\ 5 & 3 \end{pmatrix} \begin{pmatrix} 2 \\ 3 \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \\ 19 \end{pmatrix}$$

$$W = (X^T X + \lambda I)^{-1} (X^T Y)$$

$$= \frac{1}{95} \begin{bmatrix} 89 & -18 & -8 \\ -18 & 41 & -24 \\ -8 & -24 & 21 \end{bmatrix} \begin{bmatrix} 5 \\ 15 \\ 19 \end{bmatrix}$$

$$= \frac{1}{95} \begin{bmatrix} 23 \\ 69 \\ -1 \end{bmatrix}$$

$$= \begin{array}{c|l} \begin{bmatrix} 0.242 \\ 0.726 \\ -0.0105 \end{bmatrix} & \begin{matrix} c \\ m_1 \\ m_2 \end{matrix} \end{array}$$

$$\boxed{\hat{y}_{pred} = c + m_1 x_1 + m_2 x_2.}$$

$$= 0.242 + 0.726 X_1 - 0.0105 X_2$$

$$= 0.242 + 0.726(3) - 0.0105(5)$$

$$\hat{y}_1 = 2.3675 \quad , \quad Y_1 = 2.$$

$$\hat{y}_2 = 2.3885 \quad , \quad Y_2 = 3.$$

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

$$= \frac{1}{2}\left((2-2.3675)^2 + (3-2.3885)^2\right).$$

$$= \frac{1}{2} \times 0.5099885.$$

$$= 0.255$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2} \qquad \hat{y} = \frac{5}{2} = 2.5.$$

$$= 1 - \frac{0.255 \times 2}{(2-2.5)^2 + (3-2.5)^2}$$

$$R^2 = -0.0426$$

$$RMSE = 0.5099$$

$$\Rightarrow \quad C = 0.254, \quad m_1 = 0.747, \quad m_2 = 0.018$$

$$\hat{y_1} \text{ pre} = 0.254 + 3(0.747 + 5(0.018)$$

$$= 2.585$$

$$\hat{y_2} \text{ pre} = 0.254 + 3(0.747) + 3(0.018)$$

$$= 2.544.$$

$$MSE = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2$$

$$= \frac{1}{2} \left[(2-2.585)^2 + (3-2.549)^2 \right]$$

$$= 0.2729$$

$$R^2 = 1 - \frac{\sum_{i=1}^{N} (y_i - \hat{y}_i)^2}{\sum_{i=1}^{N} (y_i - \bar{y})^2}$$

$$= 1 - \frac{0.5457}{0.5}$$

$$= -0.0915$$

$$RMSE = 0.522399$$